

Koros Emmanuel Kipsang

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Portfolio: <https://korosemmanuel.github.io/Koros-Portfolio/>

PROFESSIONAL PROFILE

A confident and pro-active individual with knowledge in geospatial information system. I believe in teamwork as well as treasure using the set goals to gauge my level of success and career development. I am a hard of hearing person who is very flexible and able to work well under all normal situations. I am also result driven with exceptional communication and interpersonal skills.

CAREER OBJECTIVE

To enhance a culture of excellence giving all I can in every working position I get involved in, always ready and willing to learn and continually building responsible and accountable personalities whenever I find myself.

To become a hardworking and committed employee and ensure continuous generation of outstanding results based on the responsibilities entrusted on me.

To enhance teamwork and cooperate with other employees to bring success to the organization/institution

ACADEMIC BACKGROUND

2019 –2024: JOMO KENYATTA UNIVERSITY OF AGRICULTURE AND TECHNOLOGY

Bachelors' Degree in Geospatial Information Science. Graduated with second class honours (Lower division).

2015 -2018: METEITEI BOYS HIGH SCHOOL

Kenya Certificate of Secondary Education (KCSE)

2010 -20014: ST JOSEPH’S ACADEMY - KITURO

Studied as from standard five and sat for the Kenya Certificate of Primary Education (**KCPE**).

2003 -2009: SIGINWO PRIMARY SCHOOL

Studied here from my kinder garden to standard four before transferring to St Joseph’s Academy - Kituro.

ADMINISTRATIVE SKILLS AND EXPERTISE

- | | | | |
|------------------------|----------------------------------|-----------------------|--------|
| - AutoCAD | - Rstudio | - R | - HTML |
| - Spatial Analysis | - Database | - Data Science | |
| - Remote Sensing | - Web Development | - Data visualization, | |
| - Python | - JavaScript (React.js, Node.js) | | |
| - Land Surveying | - ArcGIS Pro | - QGIS, | |
| - Google Earth Engine | - Google Cloud | - GeoServer, | |
| - PostgreSQL/PostGIS | - MongoDB | - Leaflet.js | |
| - KoboCollect / ODK | - Engineering Survey | | |
| - GPS & Mobile Mapping | | | |

PROFESSIONAL EXPERIENCE

Freelance

Feb 2024 to Date

GIS & Remote Sensing

- Processed and analyzed drone, UAV, and satellite imagery using various GIS tools for land cover classification, change detection, suitability analysis, and 3D terrain modeling.
- Generated DEMs, DSMs, contours, and slope maps to support land management, infrastructure planning, and environmental assessments.
- Processed collected GPS data to produce topographical maps.
- Conducted hydrological modeling using the SWAT model to evaluate the impacts of land use change on watershed behavior and water availability.
- Performed coordinate transformations to ensure survey data met project accuracy standards.
- Designed and implemented Spatial Data Infrastructures (SDI) using PostgreSQL/PostGIS and GeoServer, enabling secure data access, WMS/WFS services, and WebGIS publication.

- Conducted multi-criteria decision analysis (MCDA) and suitability modeling for site selection and resource allocation.
- Digitized and georeferenced survey maps for multiple areas within Nandi County, ensuring accurate spatial alignment with national coordinate systems and GIS databases.
- Conducted image classification (supervised and unsupervised) for forest cover, land degradation, and vegetation health analysis.
- Performed NDVI, LULC, and change detection analyses to support environmental monitoring and conservation planning.
- Digitized and updated spatial features such as roads, parcels, rivers, and utilities from satellite imagery and survey data.

GIS Enumerator / Mapping Assistant (Contract)

Renaissance Company – Uasin Gishu County Jan 2026 – Feb 2026 (10 days field exercise)

- Participated in a large-scale spatial data collection and mapping exercise across Uasin Gishu County under tight timelines.
- Conducted field data collection using Android mobile devices, GPS-enabled tools, and digital data collection platforms such as KoboCollect / ODK.
- Captured and validated spatial features including roads, land parcels, public facilities, and points of interest based on predefined mapping criteria.
- Applied map reading and spatial interpretation skills to ensure positional accuracy and completeness of collected data.
- Performed data quality checks, including verification of coordinates, attributes, and completeness before submission.
- Worked collaboratively with supervisors and team members to meet daily targets and ensure timely completion of the assignment.
- Gained hands-on experience in field-based GIS workflows, data management, and real-world spatial data challenges.

Data Analyst / Data Scientist (Spatial Focus)

- Cleaned, transformed, and analyzed spatial and non-spatial datasets using Python (Pandas, NumPy, GeoPandas) and R.
- Built predictive and statistical models for environmental risk assessment, agricultural yield estimation, and resource planning.
- Applied machine learning techniques (classification and regression) to spatial datasets for trend detection and forecasting.
- Designed interactive dashboards and data visualizations using Python and web-based tools to communicate insights effectively.

Web GIS Developer

- Developed and deployed GIS-enabled web applications using React.js, Node.js, PostgreSQL/PostGIS, MongoDB, and Leaflet/Mapbox.
- Published spatial data services and built interactive mapping platforms for real-time visualization and querying.
- Implemented role-based authentication and secure APIs for managing sensitive spatial datasets.
- Optimized web applications for performance, scalability, and cross-device compatibility.

Internship

Jan 2023 - April 2023

Kenya Forest Research Institute

- Conducted hydrological analysis and land suitability modeling to support environmental management initiatives.
- Collected, processed, and mapped GPS field data for forest and land-use assessments.
- Performed terrain analysis, image classification, and spatial modeling using advanced GIS tools.
- Analyzed collected data to produce maps and spatial statistics for the research project.

TECHNICAL PROFECIENCY

- **Operating Systems:** Windows XP, 7, 8.1, 10
- **Other:** Machine learning Algorithms, Artificial Intelligence

REFERENCES

Dr. Mercy Mwaniki

Chairperson,

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